Question 1

a)

The disjoint cycle decomposition of f is found by reading it off its matrix notation:

$$f = (1\ 2\ 3)(4\ 5\ 6\ 8)(7)$$

b)

By Definition 2.3.9, the cycle type of f is: $(1, 0, 1, 1, 0, 0, 0, 0)$

c)

As done in the course notes, we find f^{-1} by swapping the rows in the matrix notation of f:

$$f^{-1} = \begin{pmatrix} 2 & 3 & 1 & 5 & 6 & 8 & 7 & 4 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 1 & 2 & 8 & 4 & 5 & 7 & 6 \end{pmatrix} \tag{1}$$

Reading off the cycles from Equation 1 (or realizing that we simply reverse all the disjoint cycles individually):

$$f^{-1} = (3\ 2\ 1)(8\ 6\ 5\ 4)(7)$$

We can now read off the cycle type (or, again, realize that we have simply reversed each disjoint cycle, preserving their lengths) as $(1, 0, 1, 1, 0, 0, 0, 0)$.

For an arbitrary $n \in \mathbb{N}$, an arbitrary element g and its inverse g^{-1} *will* have the same cycle type and order. If we write out the disjoint cycle decomposition of g and g^{-1}:

$$\begin{aligned} g &= c_1 \circ \ldots \circ c_m \\ g^{-1} &= c_m^{-1} \circ \ldots \circ c_1^{-1} \end{aligned}$$

We see that g^{-1} simply consists of the inverses of the cycles of g. An inverted cycle c^{-1} has the same length as c, and therefore g and g^{-1} will have the same cycle type. Since order is a function of the cycle lengths (Proposition 2.3.12), they will also have the same order.

Question 2

a)

For convenience define
$$f := g \circ h \circ g^{-1}.$$
Now, let $M := \{x \in \mathbb{Z}_n : h \circ g^{-1}x = g^{-1}x\}$, and take $x \in M$ chosen arbitrarily. Then $fx = g \circ h \circ g^{-1}x = g \circ g^{-1}x = x$, so x is fixed by f. Since g is bijective and h is an m-cycle, we have $|M| = n - m$.

Now we investigate what happens to the m elements in $\mathbb{Z}_n \backslash M$. First, denote $h = (b_0 b_1 \dots b_m)$, and define $x_0 = gb_0$. Note that since b_0 is not fixed by h, $x_0 \notin M$. Then for $i > 0$ define $x_i = f^i x_0$. From this definition it follows that

$$x_i = f^i x_0 = \underbrace{g \circ h \circ g^{-1} \circ \cdots \circ g \circ h \circ g^{-1}}_{i \text{ times}} x_0 = g \circ h^i \circ g^{-1} x_0 = g \circ h^i b_0 = gb_i$$

and $x_m = f^m x_0 = g \circ h^m b_0 = gb_0 = x_0$. Since h is an m-cycle $b_i \neq b_j$ for all $i \neq j$, so from bijectivity of g, we have $x_i \neq x_j$ for all $i \neq j$. Thus, we have m mutually distinct elements that satisfy $x_i = fx_{i-1}$ and $x_m = x_0$. The rest of the elements in $\mathbb{Z}_n$ are fixed by f, so $g \circ h \circ g^{-1}$ is an m-cycle.

b)

Let $h = c_1 \circ c_2 \circ \cdots \circ c_k$ be an arbitrary permutation that is a disjoint composition of cycles $c_1, c_2, \dots, c_k$. Then if we 'squeeze in' $g^{-1} \circ g$ between each cycle, we get

$$\begin{aligned} g \circ h \circ g^{-1} &= g \circ c_1 \circ c_2 \circ \cdots \circ c_k \circ g^{-1} \\ &= (g \circ c_1 \circ g^{-1}) \circ (g \circ c_2 \circ g^{-1}) \circ \cdots \circ (g \circ c_k \circ g^{-1}) \end{aligned}$$

From a, this is a composition of k cycles where the length of c_j is the same as that of $g \circ c_j \circ g^{-1}$ for all $j = 1 \dots k$. Since $c_1, c_2, \dots, c_k$ are mutually disjoint, so are $(g \circ c_1 \circ g^{-1}), (g \circ c_2 \circ g^{-1}), \dots, (g \circ c_k \circ g^{-1})$. This is because, for all $x \in \mathbb{Z}_n$ there exist at most one cycle c_j that satisfy $c_j \circ g^{-1}x \neq g^{-1}x$ or equivalently $g \circ c_j \circ g^{-1}x \neq x$. Hence, h and $g \circ h \circ g^{-1}$ have the same cycle types.

c)

Assume $h \sim k$. Then $\exists g : k = g \circ h \circ g^{-1}$, which means that $g^{-1} \circ k \circ g = h$, so also $k \sim h$.

Clearly $h \sim h$ as we can just use $g = \text{id}$.

Finally, assume $h \sim k$ and $k \sim f$. Then $\exists g : k = g \circ h \circ g^{-1}$ and $\exists w : k = w \circ h \circ w^{-1}$. This means that

$$h = g^{-1} \circ k \circ g = g^{-1} \circ w \circ h \circ w^{-1} \circ g.$$

with $q = g^{-1} \circ w$ we have $q^{-1} = w^{-1} \circ g$ so $h = q \circ f \circ q^{-1}$ so $f \sim h$.

Hence, symmetry, reflexivity and transitivity are satisfied so $\sim$ is an equivalenc relation.

Question 3

a)

For $z \in \mathbb{C}$ and $n \in \mathbb{Z}_{>0}$, $z^n = 1$ implies that $|z| = 1$, and $Arg(z) = \frac{2\pi k}{n}$, where $k \in \{0, \ldots, n-1\}$. We will write $\frac{k}{n}$ as w, and observe that $w \in \mathbb{Q}$.

Now let $a, b \in G$. Then the magnitude of their product $|ab| = |a||b| = 1$, and $|(ab)^n| = |ab|^n = 1^n = 1$. Similarly, for the argument:

$$\begin{aligned} Arg(ab) &= Arg(a) + Arg(b) \\ &= 2\pi w_a + 2\pi w_b \\ &= 2\pi(w_a + w_b) \end{aligned}$$

Then $w_{ab} = w_a + w_b$ is a rational number, and can be written as $w_{ab} = \frac{k}{n}$ for some k and n. Then, $Arg((ab)^n) = Arg(ab)n = 2\pi k$. An argument that is an integer multiple of 2π, and a magnitude of 1, means that $(ab)^n = 1$, and that therefore $ab \in G$. Multiplication is therefore a law of composition on G.

Multiplication in $\mathbb{C}$ is associative, and $1 \in G$ is the identity element, since it is the multiplicative identity of $\mathbb{C}$.

For inverses, recall that $a \in G$ implies $a^n = 1$ for some n. Therefore, $a^{n-1}a = aa^{n-1} = 1$, and a^{n-1} is therefore the inverse of a. Since multiplication is a law of composition on G, we also have that $a^{n-1} \in G$. Then a has an inverse in G.

Since

1. multiplication is a law of composition on G,
2. multiplication in $\mathbb{C}$ is associative,
3. there exists an identity element $1 \in G$,
4. every element in G has an inverse,

G is a group under multiplication.

b)

No, $1 \in G$, and $1 + 1 = 2 \notin G$.